

Engineering Ethics

“Ethics in the Innovation Process”

Hochschule Darmstadt University of Applied Sciences

May 08, 2023

Agenda for today

- Introduction
- Innovation Processes
- Shaping of technologies
- Evaluating technologies
- Values and Ethical Concerns
- Case No. 2

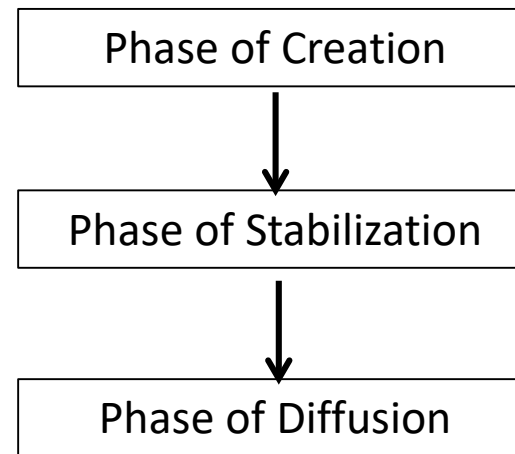
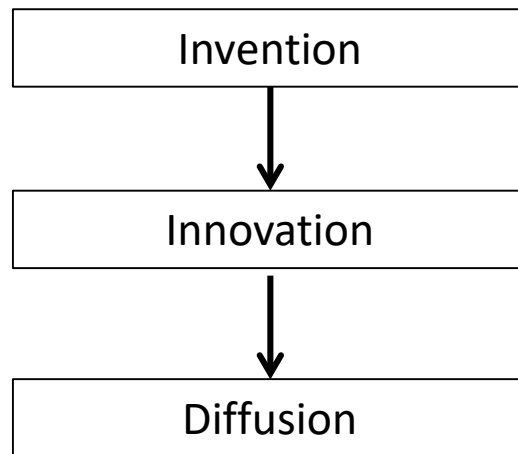
Introduction

- Any questions
- Exam, will be a 60-90 minute test on

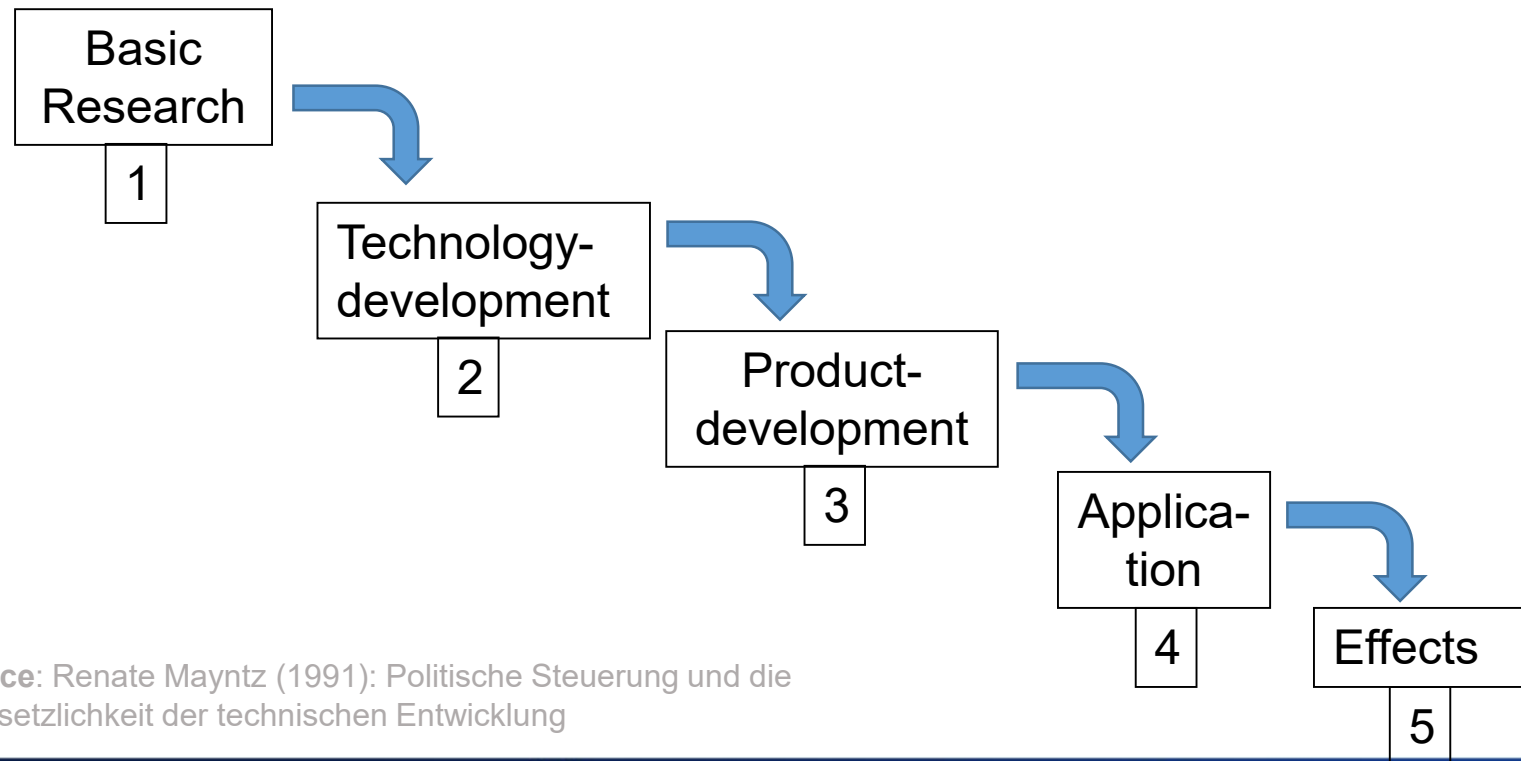
Monday June 26 – 12:30 – 14:00 h

Innovation Processes

... from the perspective of a single development project

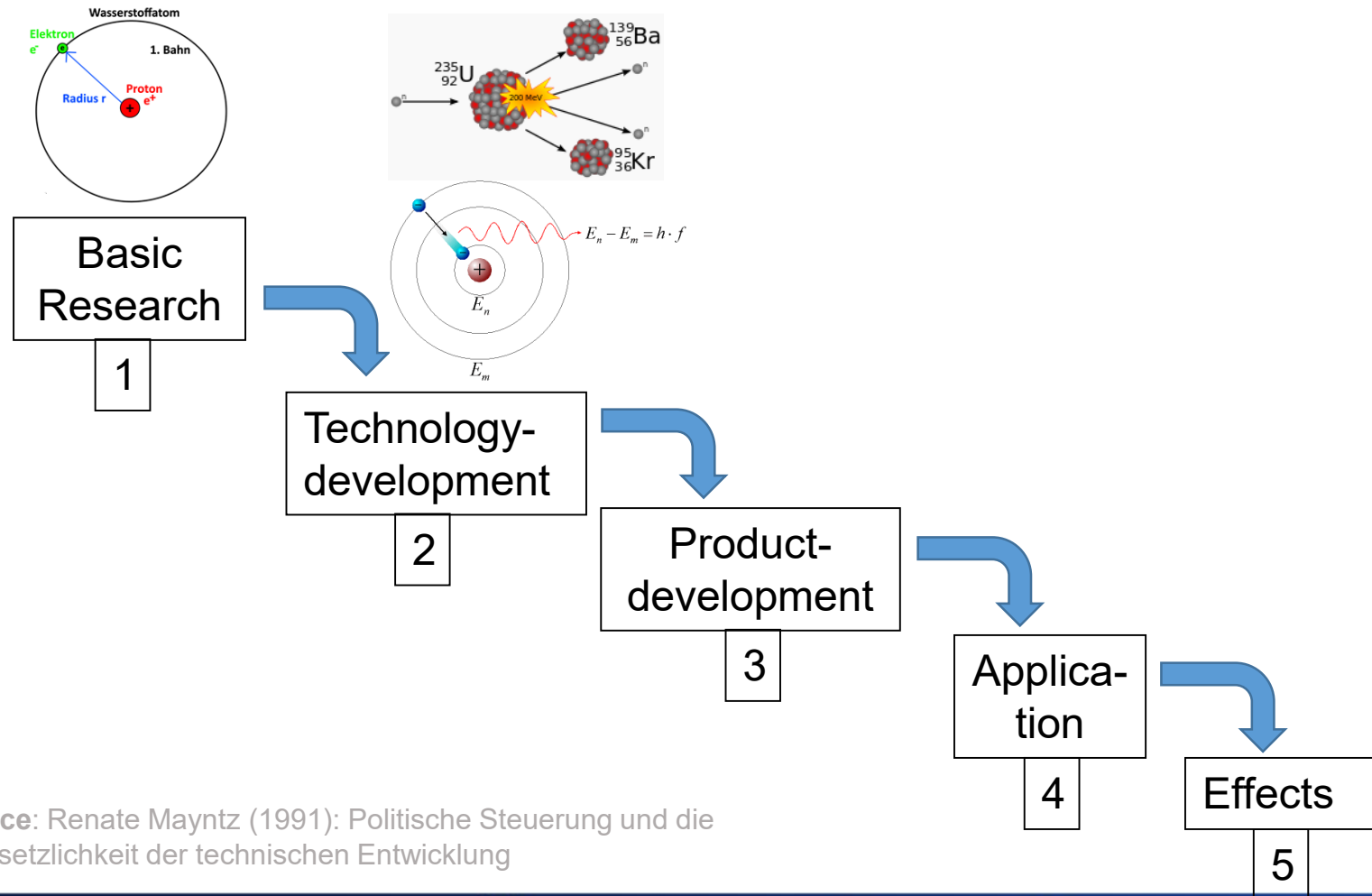


A phase scheme of technical development



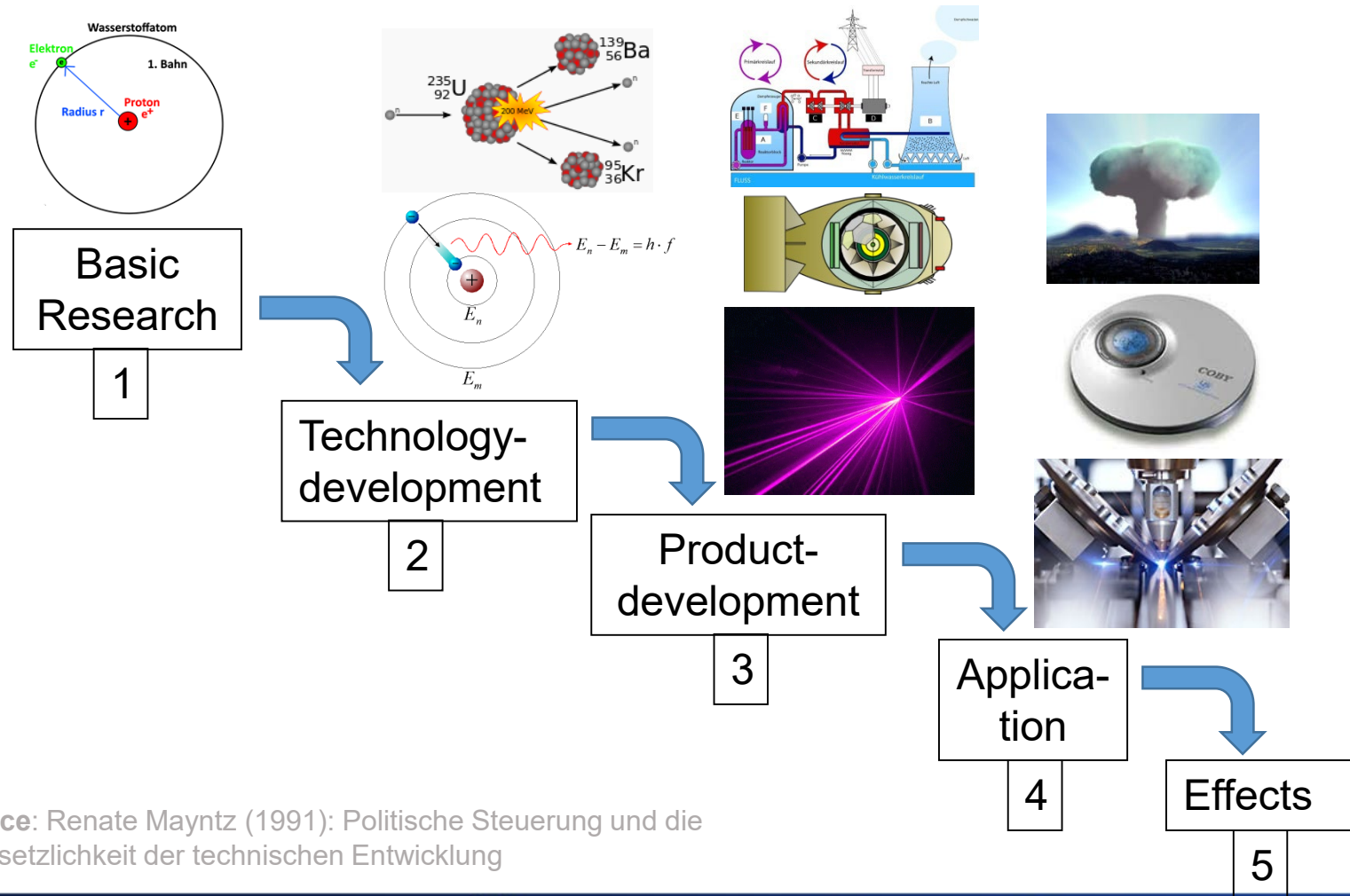
Reference: Renate Mayntz (1991): Politische Steuerung und die Eigengesetzlichkeit der technischen Entwicklung

A phase scheme of technical development



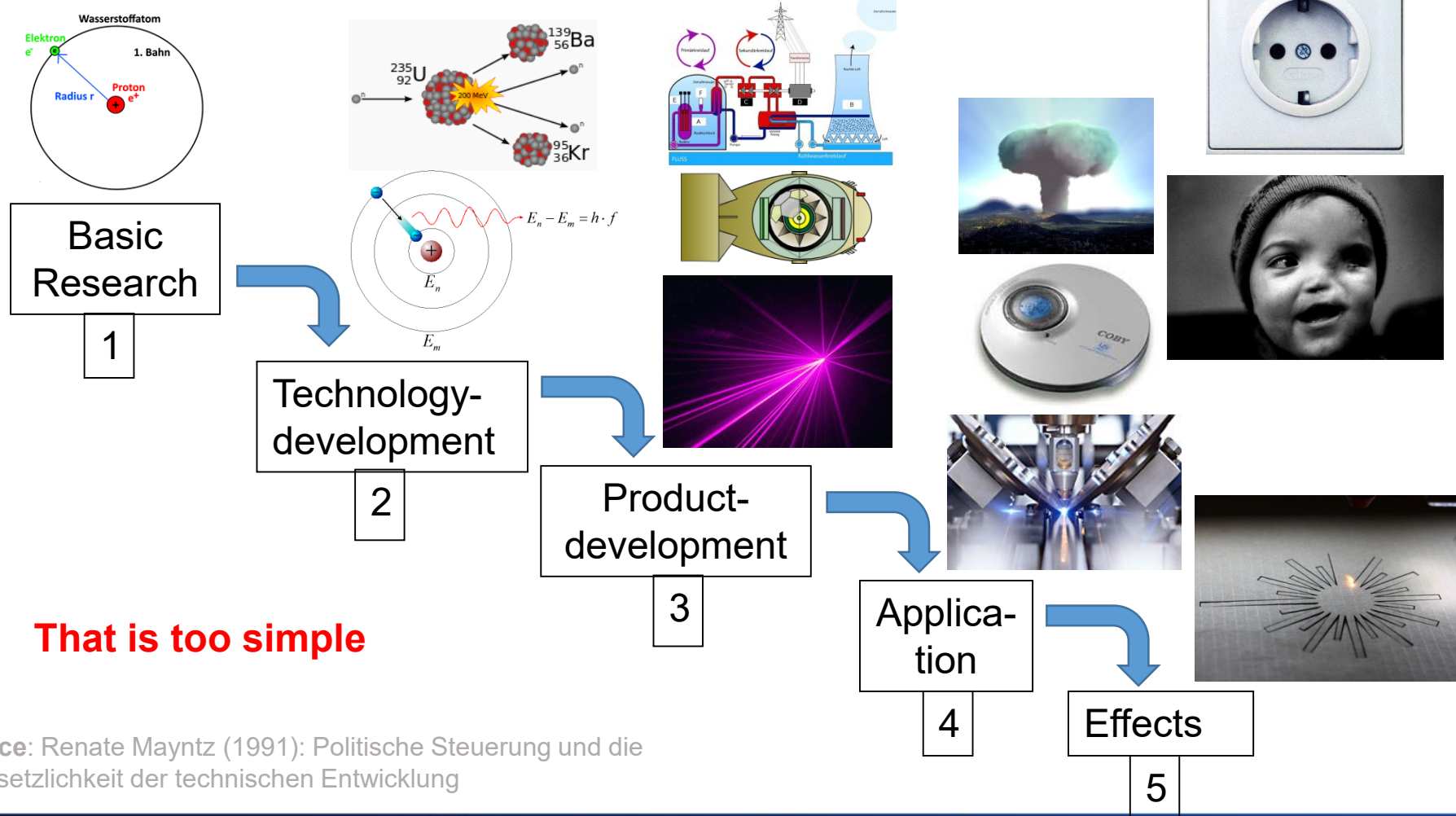
Reference: Renate Mayntz (1991): Politische Steuerung und die Eigengesetzlichkeit der technischen Entwicklung

A phase scheme of technical development

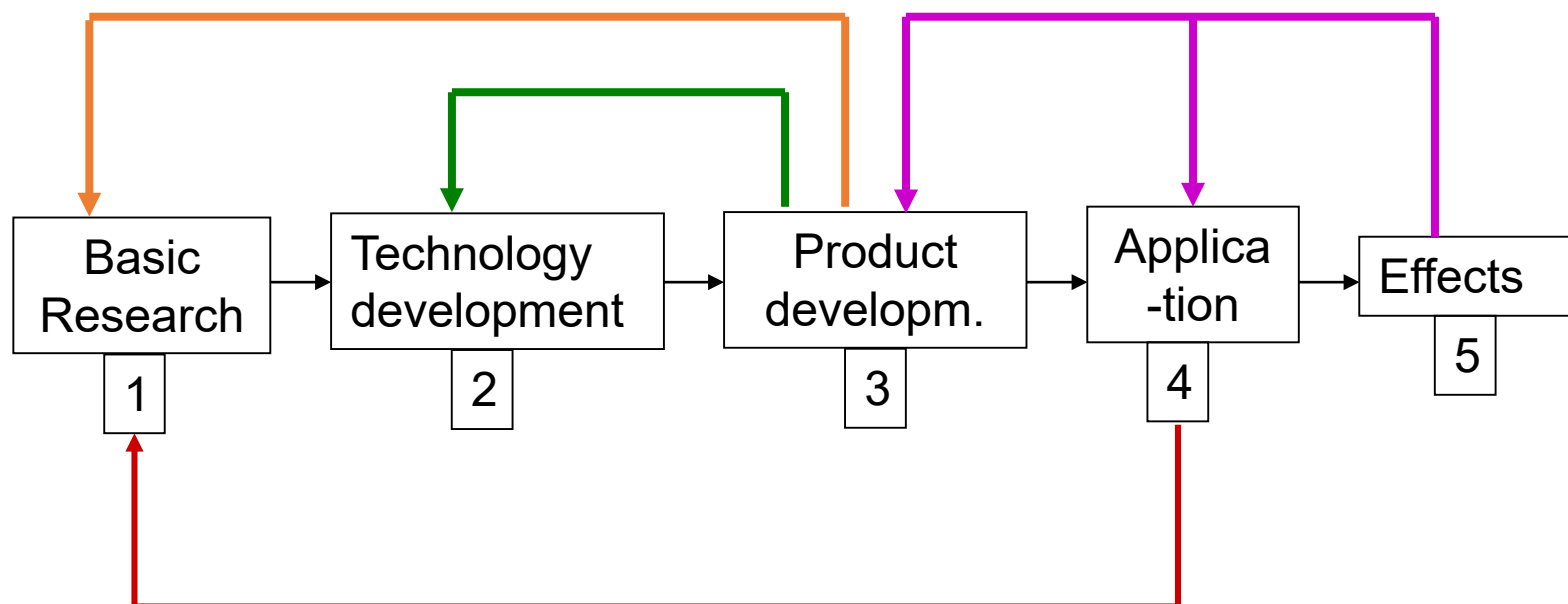


Reference: Renate Mayntz (1991): Politische Steuerung und die Eigengesetzlichkeit der technischen Entwicklung

A phase scheme of technical development

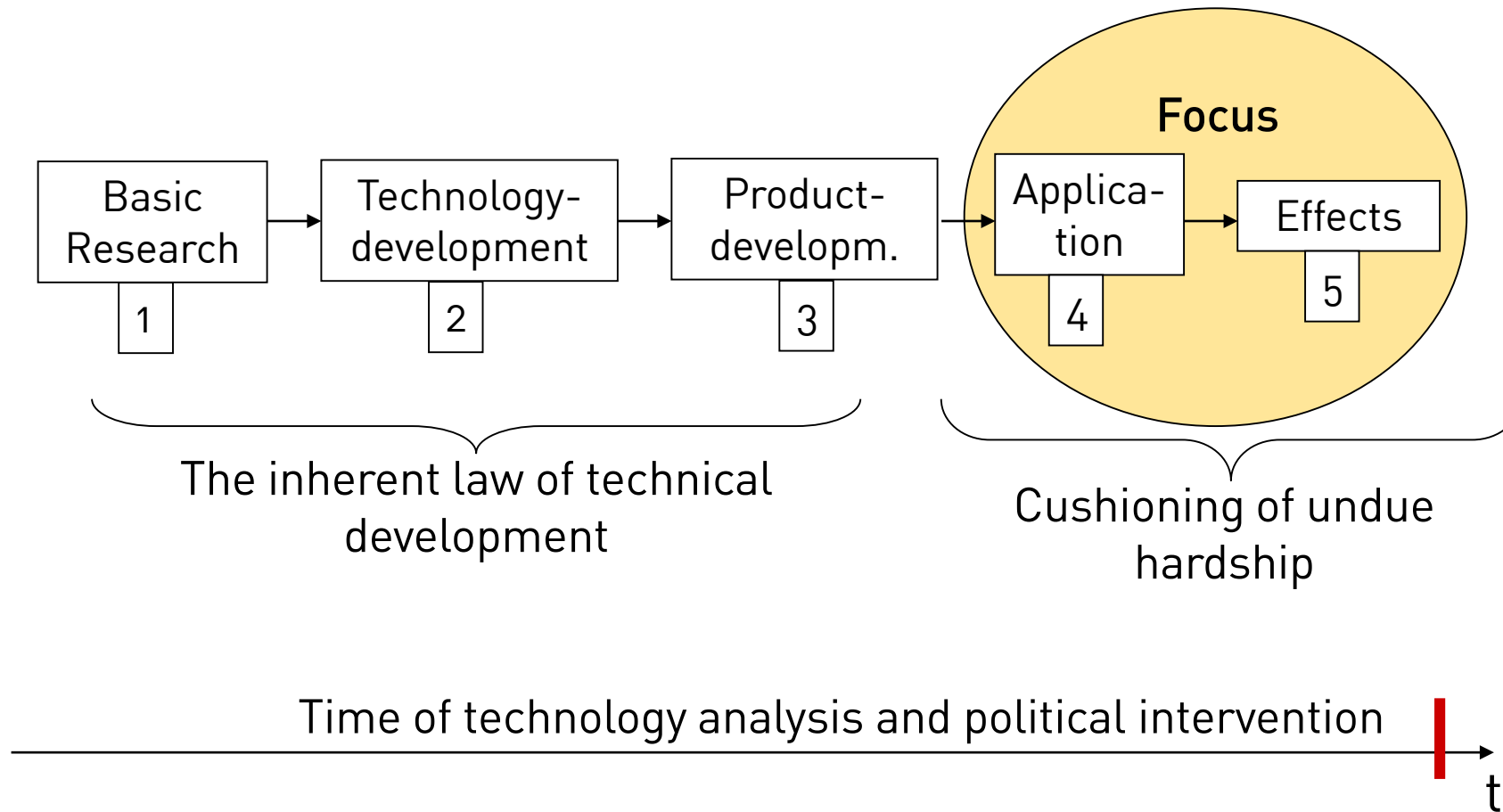


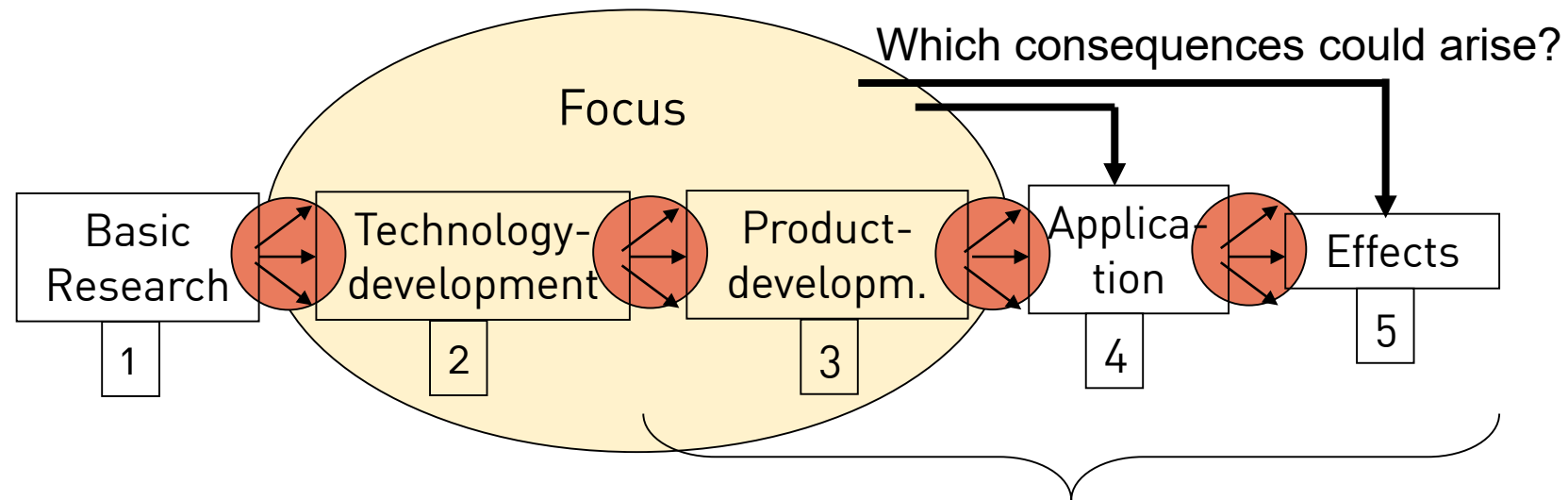
Reference: Renate Mayntz (1991): Politische Steuerung und die Eigengesetzlichkeit der technischen Entwicklung



Example:
Laser material processing





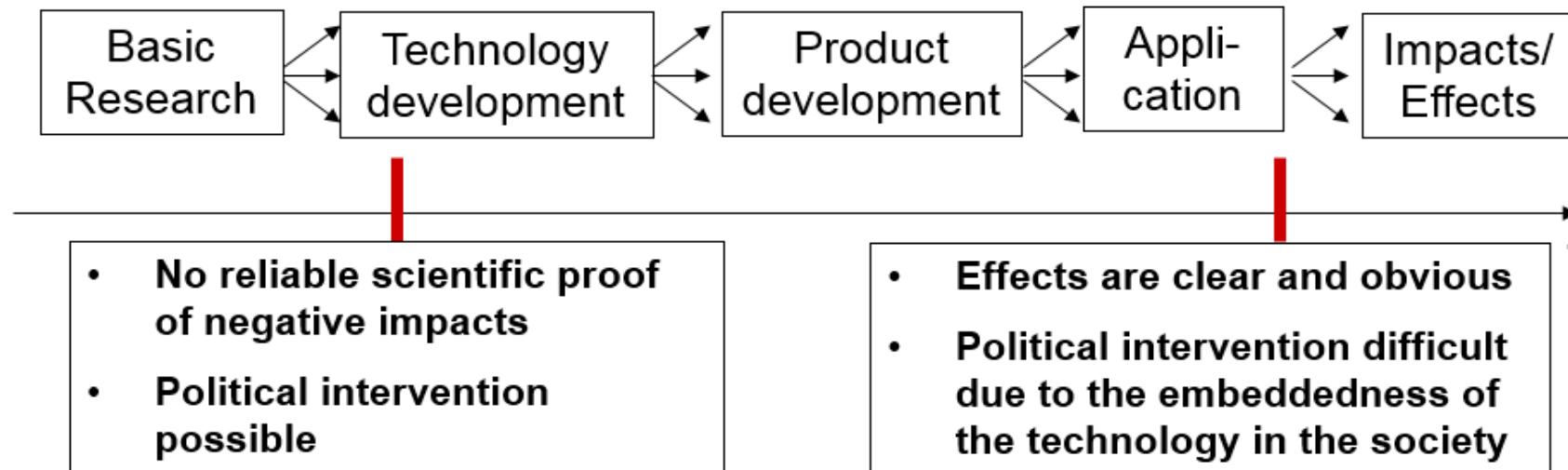


Action? or Behaviour?

- Intention is the main difference:
 - The firm/research institution/??? in focus **wants** to act in a certain way.
 - It is „planned“
 - When acting, the individual activities take place consciously
- In an organizational context you are always here !

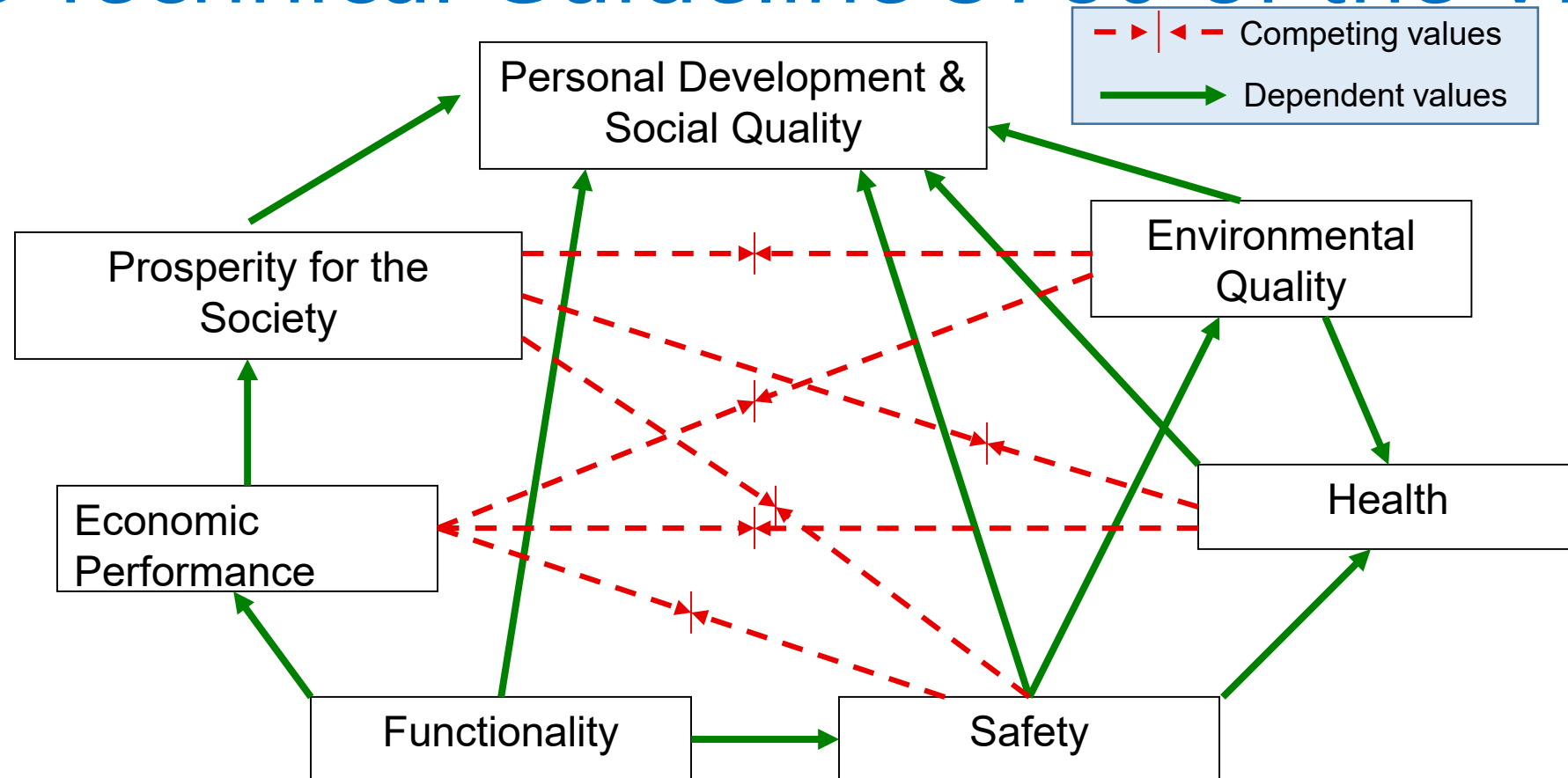


The Collingridge Dilemma



Collingridge, D. (1980). The social control of technology. St. Martin. New York.

The Technical Guideline 3780 of the VDI



Values and potential ethical concerns

- Acceptance
- Workplace effects
- Quality of work and job satisfaction
- Occupational health and safety
- (Preservation of) Biodiversity
- Data protection and security
- (Preservation of) democracy
- Energy saving
- Ergonomic design
- Human compatibility
- Cultural compatibility
- (Reduction of) noise pollution
- (Preservation of) Livelihood
- Personal development
- Conservation of resources
- Social compatibility
- Environmental compatibility
- Environmental quality
- Consumer protection
- (social) prosperity
- (social) desirability
- Social compatibility
- Demand adequacy
- Sustainability
- Participation
- Ethical criteria
- Error-friendliness
- Common good
- Gender criteria
- Health
- Product liability
- Skills development
- Recyclability
- • • •